

## SIX



## SKEWNESS AND ASYMMETRY

*We introduce the concept of skewness: why the terms “bull” and “bear” have limited meaning outside of zoology. A vicious child wrecks the structure of randomness. An introduction to the problem of epistemological opacity. The penultimate step before the problem of induction.*

*The Median Is Not the Message*

The writer and scientist Steven Jay Gould (who, for a while, was my role model), was once diagnosed with a deadly form of cancer of the lining of the stomach. The first piece of information he received about his odds of making it was that the *median* survival for the ailment is approximately eight months; information he felt akin to Isaiah's injunction to King Hezekiah to put his house in order in preparation for death.

Now a medical diagnosis, particularly one of such severity, can motivate people to do intensive research, particularly those prolific writers like Gould who needed more time with us to complete a few book projects. The further research by Gould uncovered a very different

story to the information he had initially been given; mainly that the *expected* (i.e., average) survival was considerably higher than eight months. It came to his notice that *expected* and *median* do not mean the same thing at all. Median means roughly that 50% of the people die before eight months and 50% survive longer than eight months. But those who survive would live considerably longer, generally going about life just like a regular person and fulfilling the average 73.4 or so years predicted by insurance mortality tables.

There is asymmetry. Those who die do so very early in the game, while those who live go on living very long. Whenever there is asymmetry in outcomes, the *average* survival has nothing to do with the *median* survival. This prompted Gould, who thus discovered the concept of skewness, to write his heartfelt piece “The Median is Not the Message”. His point is that the concept of median used in medical research does not characterize a probability distribution.

I will simplify Gould's point by introducing the concept of *mean* (also called *expectation*) as follows by using a less morbid example, that of gambling. I will give an example of both asymmetric odds and asymmetric outcomes to explain the point. Asymmetric odds means that probabilities are not 50% for each event, but that the probability on one side is higher than the probability on the other. Asymmetric outcomes mean that the payoffs are not equal.

Assume I engage in a gambling strategy that has 999 chances in 1,000 of making \$1 (event A) and 1 chance in 1,000 of losing \$10,000 (event B), as in Table 6.1.

Table 6.1

| <i>Event</i> | <i>Probability</i> | <i>Outcome</i> | <i>Expectation</i> |
|--------------|--------------------|----------------|--------------------|
| A            | 999/1000           | \$1            | \$.999             |
| B            | 1/1000             | –\$10,000      | –\$10.00           |
|              |                    | Total          | –\$9.001           |

My expectation is a loss of close to \$9 (obtained by multiplying the probabilities by the corresponding outcomes). The *frequency* or *probability* of the loss, in and by itself, is totally irrelevant; it needs to be judged in connection with the *magnitude* of the outcome. Here A is